

Vehicle-Level Control Concept for 3-Wheeler Auto-Rickshaw with Independent Rear Hub Motors

Proposed Solution: Active Electronic Differential with Robust Stability Control & Regenerative Vectoring.

1. System Architecture & Sensor Suite

To replace the mechanical differential while ensuring safety under variable commercial payloads, we propose a **Hierarchical Control Architecture**. A central Vehicle Control Unit (VCU) handles dynamics planning, while distributed Motor Controllers execute high-frequency traction tasks.

- **Driver Input Sensors:**
 - **Steering Angle Sensor (SAS):** Mounted on the column. Outputs Steering Angle (δ) for trajectory planning.
 - **Pedal Position Sensors:** Maps driver intent to Total Torque Request (T_{total}). Aggregates positive throttle ($+T_{req}$) and negative regenerative braking ($-T_{req}$) demands.
- **Vehicle State Sensors:**
 - **Inertial Measurement Unit (IMU):** A **Bosch BMI088** (6-axis) mounted at the chassis Center of Gravity (CG). Outputs Yaw Rate (ω_{actual}) and Lateral Acceleration (a_y) for stability and rollover detection.
 - **Wheel Speed Sensors:** High-resolution encoders on each hub. Output Wheel Speed (ω_{wheel}) for real-time Slip Ratio (λ) calculation.

2. Conceptual Control Model (Block Diagram)

The system processes inputs through three logical layers to generate precise motor commands.

[INSERT BLOCK DIAGRAM IMAGE HERE]

(Caption: Figure 1: Distributed Control Architecture featuring Reference Model, SMC Stability Layer, and Local Slip Regulation Loops.)

3. Hierarchical Control Strategy

Layer 1: Reference Model (Kinematic Intent)

Translates driver inputs into an "Ideal Vehicle State" using Ackermann steering geometry.

- **Function:** Calculates the Target Yaw Rate (ω_{des}) based on a 2-DOF bicycle model.
 - *Equation:* Target Yaw Rate $\approx$ (Velocity $\times$ Steering Angle) / Wheelbase
- **Output:** Sends Target Yaw Rate to the Upper Controller.

Layer 2: Upper-Level Controller (Stability & Safety)

Acting as the "Stability Brain," this layer compares the driver's intent (ω_{des}) with the vehicle's reality (ω_{actual}).

- **Robustness via Sliding Mode Control (SMC):** Unlike PID controllers which require precise tuning for fixed masses, we utilize **SMC** with a discontinuous switching gain. This ensures stability despite the massive mass variance (Moment of Inertia changes) typical of commercial goods carriers.
- **Rollover Prevention (Safety Override):** Simultaneously monitors the **Lateral Load Transfer Ratio (LTR)** derived from Lateral Acceleration.
 - *Normal Mode:* Outputs a Corrective Yaw Moment (M_z) to reduce steering error.

- *Danger Mode ($LTR > 0.8$):* If the inner wheel is at risk of lifting, the controller overrides steering logic and commands a braking moment to stabilize the chassis.

Layer 3: Torque Allocator (The "Virtual Differential")

Converts the calculated Yaw Moment (M_z) into specific torque commands for the left and right motors.

- **Logic:** It superimposes the steering effort onto the driver's requested torque (T_{total}).
 - *Right Motor Torque:* $(Total\ Torque / 2) + Adjustment\ Torque$
 - *Left Motor Torque:* $(Total\ Torque / 2) - Adjustment\ Torque$
 - *Where:* $Adjustment\ Torque = (Yaw\ Moment \times Tire\ Radius) / Track\ Width$
- **Note:** Using Track Width in the denominator ensures the couple force correctly generates the required Yaw Moment.

Block D: Local Slip Control Loop (Traction Execution)

Embedded directly in the Motor Controllers (FOC), this high-frequency loop ensures traction.

- **Function:** It continuously monitors the Slip Ratio (λ) by comparing wheel speed to vehicle speed.
- **Action:** It acts as a "Ceiling Limiter." If Slip Ratio $> 15\%$ (loss of traction), it overrides the VCU torque command and cuts current locally to re-adhere the tire, ensuring maximum acceleration on low-friction surfaces without VCU latency.

Qualitative Analysis & Design Feasibility

1. Longitudinal Performance: Active Traction Optimization

Unlike a mechanical open differential, which sends equal torque to both wheels regardless of grip (often causing one wheel to spin helplessly on low-friction surfaces), our **Block D Slip Limiter** ensures continuous acceleration.

- **Launch & Acceleration:** By continuously monitoring the Slip Ratio, the system maintains wheel slip within the optimum window of **10-15%**. This is the physical peak of the friction curve where the tire generates maximum longitudinal force.
- **Surface Independence:** On split-friction surfaces (e.g., one wheel on tarmac, one on dirt), the controller independently throttles the slipping wheel while maintaining full torque on the gripping wheel.
- **Combined-Slip Braking Stability (Electronic EBD):** During "Braking-While-Turning" maneuvers, the system acts as an active **Electronic Brake Force Distributor**.
 - *Vectoring:* The Allocator shifts regenerative load to the inner wheel ($Left\ Torque = Total / 2 - Adjustment$) to create a pivoting moment, helping the vehicle turn in while slowing down.
 - *Anti-Lock Logic:* Simultaneously, the Slip Control Loop prevents wheel lock-up. If the vectoring demand exceeds the tire's friction circle (causing a slide), the controller instantly reduces the regenerative torque to restore lateral steering authority.

[INSERT SLIP VS FRICTION GRAPH HERE]

(Caption: Figure 2: The Controller targets the green shaded region (10-15% Slip) for maximum traction.)

2. Lateral Stability & Steering Behavior

The **Layer 2 Sliding Mode Controller (SMC)** actively manages the vehicle’s yaw dynamics to solve the inherent instability of three-wheelers.

- **Turn-In Response:** Upon entering a turn, the **Torque Allocator** instantly generates a positive Yaw Moment (+M_z) by over-driving the outer wheel. This reduces the steering effort required by the driver and eliminates the "understeer/plowing" tendency typical of rigid-axle vehicles.
- **Rollover Prevention (The Safety Override):** The system continuously calculates the Lateral Load Transfer Ratio (LTR).
 - *Threshold:* If LTR exceeds **0.8** (indicating imminent inner-wheel lift), the SMC transitions to a safety state.
 - *Action:* It commands a massive negative braking torque on the outer wheel. This creates a stabilizing moment that realigns the chassis and reduces Lateral Acceleration (a_y) before a rollover can occur.

[INSERT STATE MACHINE DIAGRAM HERE]

(Caption: Figure 3: Safety logic transitions to rollover prevention when LTR exceeds critical threshold.)

3. Design Constraints & Feasibility Analysis

Our proposed architecture directly addresses the operational constraints of a commercial mass-market vehicle:

Constraint	Challenge	Our Control Solution
Mass Variation	Payload changes (Empty vs. Fully Loaded) alter dynamics, causing PID loops to fail.	SMC (Layer 2) utilizes a "Switching Gain" tuned for maximum inertia. This provides inherent robustness against 30-40% mass changes without complex adaptive tuning.
Sensor Cost	Direct sideslip and torque sensors are too expensive for this segment.	Kinematic Observation: We estimate vehicle states using sensor fusion from the low-cost Bosch BMI088 IMU and standard Hall-effect wheel encoders.
Efficiency	Stability systems (ESP) typically waste energy by braking.	Regenerative Vectoring: The Torque Allocator prioritizes regenerative braking on the inner wheel during corrections, recovering kinetic energy while stabilizing the turn.